# Supplementary Material for
# Auto-Decte: Trust-Constrained Document Intelligence

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## 1 Artifact inventory

The package contains raw cell and whole-form outcomes, complete workflow traces, trust-boundary tests, resilience timings, generated plots and tables, and a manifest with SHA-256 hashes. The recorded run used source commit `195355a` with a clean tracked source state and produced 27 hashed outputs. The authority-integration benchmark artifacts (benchmarks/pipeline_authority_benchmark.py) are generated from the integrated source commit `68f7b93` with a fixed injected clock and seed; two independent runs produced byte-identical pipeline_authority.json and manifest.json files whose artifact hashes were independently recomputed.

| Artifact | Purpose |
|---|---|
| `artifacts/recognition_raw.json` | 13,050 model–case rows |
| `artifacts/recognition_summary.json` | thresholds, intervals, baselines, robustness |
| `artifacts/form_raw.json` | 1,200 form–condition–variant rows |
| `artifacts/form_summary.json` | whole-form metrics and component-suppression outcomes |
| `artifacts/form_workflow.json` | 60 import-to-export workflow outcomes |
| `artifacts/trust_faults.json` | six controlled fault outcomes |
| `artifacts/trust_ablation.json` | candidate-isolation intervention |
| `artifacts/trust_stress.json` | 500 randomized fault trials |
| `artifacts/resilience.json` | checks and latency repetitions |
| `artifacts/manifest.json` | hashes, environment, commands, source identity |

## 2 Cell-level protocol

Calibration uses seeds 101 and 202 (1,740 rows). Evaluation uses seeds 303, 404, and 505 (2,610 rows). The evaluation rows comprise 29 conditions applied to 90 seed–font–digit base units. HOG–SVM training uses eight separate seeds. Thresholds from 0 to 0.10 in increments of 0.005 are evaluated; each operating threshold is frozen from the calibration-risk target before evaluation.

Cluster-bootstrap intervals use 2,000 replicates. Each replicate samples 90 base-unit identifiers with replacement and retains all condition rows belonging to each sampled unit. This preserves within-unit dependence. The descriptive Wilson interval in the generated JSON treats rows as independent and is not used as inferential evidence in the manuscript.

## 3 Perturbation-level template-matcher outcomes

Table 1 reports the complete template-matcher aggregation at $\tau = 0.08$. Each row applies one condition to the same 90 evaluation base units. Accepted accuracy is conditional on acceptance and must be interpreted with coverage.

## 4 Whole-form evaluation

Two templates and 30 evaluation seeds produced 60 base forms. Each was evaluated under clean, Gaussian-blur, rotation, perspective, and JPEG conditions, yielding 300 condition rows per variant. Table 2 reports the aggregate component and complete-form outcomes.

The complete pipeline succeeded on every clean, perspective, and JPEG row. Blur prevented QR identification while preserving alignment and field recognition. Rotation prevented marker alignment and

Table 1: Coverage and conditional accepted accuracy by perturbation.

| Condition | Severity | Coverage (%) | Accepted accuracy (%) |
|---|---|---|---|
| clean | 0.00 | 13.3 | 100.0 |
| gaussian_blur | 0.25 | 11.1 | 100.0 |
| gaussian_blur | 0.50 | 14.4 | 100.0 |
| gaussian_blur | 0.75 | 14.4 | 100.0 |
| gaussian_blur | 1.00 | 15.6 | 100.0 |
| gaussian_noise | 0.25 | 12.2 | 100.0 |
| gaussian_noise | 0.50 | 11.1 | 100.0 |
| gaussian_noise | 0.75 | 12.2 | 100.0 |
| gaussian_noise | 1.00 | 12.2 | 100.0 |
| brightness | 0.25 | 13.3 | 100.0 |
| brightness | 0.50 | 12.2 | 100.0 |
| brightness | 0.75 | 12.2 | 100.0 |
| brightness | 1.00 | 12.2 | 100.0 |
| rotation | 0.25 | 12.2 | 100.0 |
| rotation | 0.50 | 13.3 | 100.0 |
| rotation | 0.75 | 12.2 | 100.0 |
| rotation | 1.00 | 10.0 | 100.0 |
| perspective | 0.25 | 8.9 | 100.0 |
| perspective | 0.50 | 7.8 | 100.0 |
| perspective | 0.75 | 4.4 | 100.0 |
| perspective | 1.00 | 3.3 | 100.0 |
| jpeg | 0.25 | 13.3 | 100.0 |
| jpeg | 0.50 | 13.3 | 100.0 |
| jpeg | 0.75 | 11.1 | 100.0 |
| jpeg | 1.00 | 13.3 | 100.0 |
| occlusion | 0.25 | 12.2 | 100.0 |
| occlusion | 0.50 | 12.2 | 100.0 |
| occlusion | 0.75 | 6.7 | 100.0 |
| occlusion | 1.00 | 3.3 | 100.0 |

Table 2: Whole-form pipeline and component-suppression analysis.

| Variant | QR (%) | Align (%) | Digit (%) | OMR (%) | Complete (%) |
|---|---|---|---|---|---|
| full_pipeline | 80.0 | 80.0 | 86.0 | 100.0 | 60.0 |
| without_qr | 0.0 | 80.0 | 86.0 | 100.0 | 0.0 |
| without_aruco | 80.0 | 0.0 | 69.4 | 99.0 | 0.0 |

reduced digit accuracy. The separate workflow artifact records 20 recognition candidates, one versioned confirmation, XLSX export, and reverse trace for each of 60 clean forms.

# 5 Fault definitions

Wrong recognition and wrong AI suggestion inject incorrect values through candidate services. Irrelevant retrieval supplies a non-matching reference. Stale human replay submits an outdated expected version. Duplicate evidence submits identical source content twice. Attempted direct fact write queries four fact-like methods on each of three machine-facing services; all 12 methods are unavailable. The candidate-isolation ablation contrasts normal candidate persistence with test-only unsafe review-service wiring.

The randomized stress test draws new values, confidence scores, crops, references, and replay identifiers from a fixed seed. It executes 100 trials for each of five families:

| Fault family | Contained | Trials |
|---|---|---|
| wrong_recognition | 100 | 100 |
| wrong_llm_suggestion | 100 | 100 |
| irrelevant_retrieval | 100 | 100 |
| stale_human_replay | 100 | 100 |
| duplicate_evidence | 100 | 100 |

## 6 Service microbenchmarks

Across 30 repetitions after five warm-ups, median fresh-container service latency was 0.300 ms (95th percentile 0.343 ms), and median reused-container latency was 0.073 ms (95th percentile 0.085 ms). These values exclude perception, UI rendering, storage-device variance, and human review.

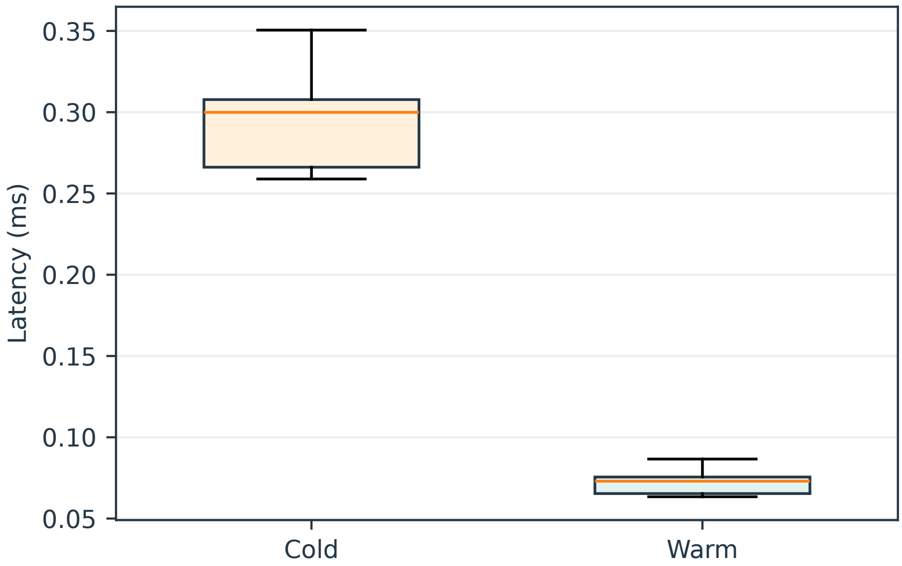


Figure 1: Application-service latency distributions.

## 7 Reproduction

From the experiment repository and its Python environment, run:

```
python -m benchmarks.run_all --config benchmarks/config/eswa-v1.json \\
  --output artifacts/eswa-journal-extension-v3
python -m benchmarks.pipeline_authority_benchmark --output \\
  <output-dir> --now 2026-08-17T12:00:00+00:00 --seed 20260817 --cases 12
uv run pytest -q
uv run ruff check .
uv run mypy
```

The recorded environment used Python 3.11.9. Tesseract was unavailable and is therefore excluded rather than simulated.